

1. INTRODUCTION

The theory of Schubert polynomials, initiated by Lascoux and Schützenberger, provides polynomial representatives for Schubert classes in the cohomology of flag varieties. In the classical setting, the Littlewood–Richardson coefficients $c_{u,v}^w$ appearing in

$$\mathfrak{S}_u(\mathbf{x}) \cdot \mathfrak{S}_v(\mathbf{x}) = \sum_w c_{u,v}^w \mathfrak{S}_w(\mathbf{x})$$

are nonnegative integers, reflecting the geometric fact that they count intersection points of Schubert varieties. Graham’s seminal theorem [?] established that in the equivariant setting these coefficients become polynomials in the negative roots x_i with nonnegative integer coefficients.

The quantum deformation of this story was developed by Kirillov and Maeno [?], who introduced quantum double Schubert polynomials $\tilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{y})$. In the quantum setting the product involves formal variables q^β encoding curve-counting information:

$$(1) \quad \tilde{\mathfrak{S}}_u(\mathbf{x}; \mathbf{y}) \cdot \tilde{\mathfrak{S}}_v(\mathbf{x}; \mathbf{z}) = \sum_d \sum_w q^d c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{z}) \tilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{z}).$$

Mihalcea [?] proved that when $\mathbf{y} = \mathbf{z}$ (the double case), the coefficients $c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{y})$ are polynomials in the negative roots $y_i - y_j$ with nonnegative integer coefficients, establishing the Peterson conjecture for equivariant quantum cohomology.

In this paper we prove the corresponding positivity for the *triple* case, where $\mathbf{y} \neq \mathbf{z}$ are independent variable sets.

The proof strategy follows the method of Mihalcea, combined with the triple Schubert calculus established by Gao and Xiong in [?]. The key idea is a reduction from the quantum triple case to the already-established double positivity of Mihalcea, via a specialization argument using the separated descents condition of [?].

2. NOTATIONS AND PRELIMINARIES

2.1. Flag variety and Weyl group. Let $G = SL_n(\mathbb{C})$ and $B \subset G$ a Borel subgroup of upper triangular matrices. The **flag variety** is $Fl_n = G/B$, a smooth complex algebraic variety of dimension $\binom{n}{2}$. Its Weyl group is the symmetric group S_n , generated by simple reflections s_i ($i = 1, \dots, n-1$) satisfying the Coxeter relations. For $w \in S_n$ the **length** $\ell(w)$ equals the number of inversions of w , and $w_0 = (n, n-1, \dots, 1)$ denotes the longest element.

The set of **descents** of w is

$$\text{Des}(w) = \{ i : w(i) > w(i+1) \}.$$

We say (u, v) satisfies the **separated descents condition** if $\max \text{Des}(u) \leq \min \text{Des}(v)$; see [?, Definition 2.1].

2.2. Divided difference operators. For $i = 1, \dots, n-1$, the **divided difference operator** ∂_i acts on polynomials in variables $x_1, \dots, x_n$ by

$$\partial_i f = \frac{f(\dots, x_i, x_{i+1}, \dots) - f(\dots, x_{i+1}, x_i, \dots)}{x_i - x_{i+1}}.$$

These operators satisfy $\partial_i^2 = 0$ and the braid relations, so for any $w \in S_n$ the operator $\partial_w = \partial_{i_1} \cdots \partial_{i_k}$ is well-defined when $(i_1, \dots, i_k)$ is a reduced word for w .